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Quantum word embedding for machine learning

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Abstract

The accelerated progress in quantum computing has enabled a new form of machine intelligence that runs on quantum hardware, which holds great promise for more powerful computational models in various learning tasks. An emergent application of Quantum Machine Intelligence (QMI) is Quantum Natural Language Processing (QNLP). This paper proposes a multi-dimensional, finite automaton model for quantum word embedding (QWE) via the Galois field. We demonstrated the model to three applications: (1) English vocabulary, (2) amino acid-based genetic codes, and (3) DNA-based genetic codes. The numerical results obtained from the proposed algorithm for the English vocabulary indicate that it produces more representative word features than Word2Vec based on the word distance metric. Second, the proposed algorithm is also utilized to model RNA-Protein interaction based on the latent distance of a given molecule, which is demonstrated on three large datasets, namely RPI369, RPI1807, and RPI2241. Finally, two embedding techniques for DNA-based genetic codes are proposed in this work: Two-state Lackadaisical Encoding (TCE) and Topological-Cyclic Encoding (TLE). These techniques enable extracting relevant features for the efficacy score of gRNAs used in the CRISPR-Cas 9 system, demonstrated on 15 datasets, compared to 12 mathematical features. We make our implementation available at https://github.com/namnguyen0510/Quantum-Embedding-of-Word/tree/main.

1. Introduction

Quantum neural networks (QNNs) are a class of machine learning (ML) models that leverage principles from quantum mechanics, quantum computing, and classical artificial neural networks (ANNs). Another way to view QML/QNN is as a machine intelligence model realized on quantum hardware instead of classical computers. Specifically, the basic computation units of classical ANNs are neurons. In contrast, QNN uses qubits or qurons [1–4], which can exist in superposition or multiple states at the same time. The input data to QNNs is encapsulated as a quantum wavefunction, probabilistic-mathematical objects before measurement but collapsed into classical information post-measurement.

Although the early literature on QML/QNNs pertains to the Noisy Intermediate-Scaled Quantum (NISQ [5]) era, this field has demonstrated a growing potential and numerous advantages over classical machine learning. Theoretical analysis [6] presents an approach to evaluating the trainability and capacity of quantum neural networks (QNNs) using Fisher Information and affective dimensions. This approach demonstrates that QNNs are more effective and efficient than comparable classical artificial neural network (ANN) models. Another theoretical-geometric test is also proposed in [7] to show quantum advantages [8] in predictive performance from a data perspective, with a proof-of-concept demonstrated on intermediate-scaled systems. Additionally, quantum embedding or quantum feature maps display considerable advantages over brief features extracted by classical ANNs, discussed in metric learning [9] and the expressiveness of model predictions [10]. From an empirical standpoint, several proofs-of-concept demonstrate the effectiveness of quantum neural networks (QNNs) and quantum machine learning (QML) in various applications, as evidenced by several studies [1, 11–17]. An intriguing idea of soft quantum neuron, that simulates biological neurons, is introduced in [18], providing a good non-linear model for classification.

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1.1. Related works

The considerable computational capability of quantum computing has captured the interest of the Natural Language Processing (NLP) community. Attention-based models have emerged as the dominant neural architecture in AI-related fields, with much recent attention directed toward quantum computing's potential to enhance these models. The QSANN algorithm [19] is the initial quantum NLP (QNLP) model that introduces a parallel encoding scheme for input sequences. Using an encoding scheme that involves multiple small subsystems, each composed of 2-4 qubits to encode words or tokens from an input sequence, the QSANN algorithm presents a practical approach that can be applied to larger datasets and implemented on NISQ devices. By introducing Gaussian projected quantum self-attention, QSANN efficiently uncovers correlations between words in a high-dimensional quantum feature space, and its self-attention coefficients are shown to focus on the most relevant words. Compared to existing QNLP methods based on syntactic analysis [20] and comparable-classical self-attention neural networks [21], QSANN outperforms them on various public datasets for text classification and numerical results suggest that QSANN is resistant to quantum noise. Another benchmark evaluation for QNLP is reported in [20], in which the QML algorithm can analyze an intermediate-scaled dataset.

Moreover, Quantum Self-Attention Network (QSAN) [22] explores the theoretical basis of QSAN, including a Quantum Self-Attention Mechanism (QSAM) that is linear and reversible. QSAM solves the space complexity problem of classical Self-Attention Mechanism (SAM) caused by quadratic complexity $\mathcal{O}(n^2)$. The study also highlights the importance of the Quantum Logic Similarity (QLS) and Quantum Bit Self-Attention Score Matrix (QBSASM) components of QSAM. QLS replaces inner product operations with logical operations, making QSAN fully deployable on quantum computers and saving quantum bits usage by avoiding numerical operations. QBSASM is a by-product generated during the evolution of QSAN, which reflects the output attention distribution in the form of a density matrix. We find that QSANN and QSAN are thoughtful approaches to reducing the complexity and run-time for QML/QNN on near-term devices. We give a brief background of attention-learning in (Appendix .1).

1.2. Limitations of the current approaches

The conventional approach to encoding categorical variables $\{c_1, \ldots, c_n\}$ for ML/DL algorithm is One-hot Encoding (OHE). Specifically, the data encoding is presented as

$$\begin{aligned} &c_1 \mapsto [1,0,0, \ldots, 0]^\top \\ &c_2 \mapsto [0,1,0, \ldots, 0]^\top \\ &c_3 \mapsto [0,0,1, \ldots, 0]^\top \\ &\cdots \\ &c_i \mapsto [0,0,\ldots,1(i^{th}\text{position}), \ldots, 0]^\top \\ &\cdots \\ &c_n \mapsto [0,\ 0,\ 0, \ldots, 1] \end{aligned} \tag{1}$$

Of note, a given word of length m spanned by n characters $\mathbb{B} = \{c_1, \ldots, c_n\}$ is presented as $\boldsymbol{H}^{\text{Real}}_{m\times n}$. One of the main differences between quantum embedding and **OHE** is that the quantum feature map $|\psi(\boldsymbol{b})\rangle$ takes complex values instead of real values.

We find that quantum feature maps $|\psi(\boldsymbol{b})\rangle$ can be defined as a *representation* of character $\boldsymbol{b}$ through quantum unitary transformation. Specifically, given a quantum system of m qubits prepared on a quantum computer, the quantum feature map of character $\boldsymbol{b}$ is given by corresponding unitary transformation parameterized by $\boldsymbol{b}$ applied on $|0\rangle^{\otimes m}$; i.e.,

$$|\psi(\boldsymbol{b})\rangle = \mathcal{U}(\boldsymbol{b})|0\rangle^{\otimes m}. \tag{2}$$

We can generalize quantum feature maps as a homomorphism $\boldsymbol{\psi}$: $G \to GL(V)$ with V as finite-dimensional Hilbert vector space. The representations of infinite Hilbert vector space are out of this study's scope. We can construct a complex-valued function from $\mathbb{Z}/n\mathbb{Z}$, given by

$$\begin{aligned} &\boldsymbol{\psi}\colon \mathbb{Z}/n\mathbb{Z} \to \mathbb{C}^* \\ &\qquad m \mapsto \psi[m] = \xi^m = e^{\frac{2\pi i m}{n}} = \cos(\frac{2\pi m}{n}) + i\sin(\frac{2\pi m}{n}) \\ &\qquad m \in \mathbb{Z},\ m \in \{0,\ 1,\ 2, \ldots, n-1\} \end{aligned} \tag{3}$$

For a corpus $\mathbb{B}$ of n characters, we can directly construct orthogonal vectors from the basis set $\{\xi^0, \xi^1, \ldots, \xi^{n-1}\}$ like OHE, given as

$$
\begin{aligned}
&c_1 \mapsto [\xi^0, 0, 0, \ldots, 0]^\top \\
&c_2 \mapsto [0, \xi^1, 0, \ldots, 0]^\top \\
&c_3 \mapsto [0,0, \xi^2, \ldots, 0]^\top \\
&\cdots \\
&c_i \mapsto [0,0, \ldots, \xi^{i-1}(i^{th}\text{position}), \ldots, 0]^\top \\
&\cdots \\
&c_n \mapsto [0, 0, 0,\ldots, \xi^{n-1}]
\end{aligned} \tag{4}
$$

of note, this trivial translation from OHE to the embedded feature presented in equation (4) does not have any quantum advantage as each complex number $z = a + bi$ is defined by two real numbers $a,\ b \in \mathbb{R}$. Thus, the embedding in equation (4) requires $2 \times n$ floating points for each basis, resulting in embedded tensor of $\boldsymbol{H}^{\text{Complex}}_{d\times n\times 2}$, which has double space complexity.

1.3. Contribution

In this study, we will propose a quantum embedding technique for three types of languages, including (1) English vocabulary, (2) Amino-acid-based (AA-based) genetic codes, and (3) DNA-based genetic codes. We summarize our contribution as follows:

- The proposed Quantum Word Embedding (QWE) of English vocabulary outperforms the classical approach Word2Vec, demonstrated in section 5.2.
- The proposed QWE for AA-based genetic codes allows predicting RNA-Protein Interaction (RPI) based on Mutual Information of derived latent tensors, quantified on intermediate-scaled datasets (RPI369, RPI1807, and RPI2241) (section 5.3).
- The proposed Topological-Cyclic Encoding (TCE) of the DNA-based genetic codes is relevant to the therapeutic score (efficacy) of input-guided RNA (gRNA) used in gene editing technology. Besides, the quantum encoding vectors are spare but representative, which reduces computational resources for information retrieval of the DNA codes (section 5.4).

2. Formalism

2.1. Finite automaton modeling

Definition 1. Binary relation between two mathematical objects $\square_A$ and $\square_B$ is defined as $\circ$, written as $\square_A \circ \square_B$. We visualize the **graphical representation** of $\square_A \circ \square_B$ as $\square_A \overset{\circ}{\underset{\circ}{\rightleftarrows}} \square_B$. Conventionally, we consider the class of (1) reflexive, (2) symmetric, (3) anti-symmetric, and (4) transitive binary relation.

Definition 2. An alphabet is a finite, nonempty set of symbols, denoted as Λ. For example,

1. English alphabet consists of 26 symbols or characters.
2. Genetic code is a biological language, which different alphabet sets can present: at DNA-level, the alphabet set is $\Lambda = \{A,\ T,\ G,\ C\}$ (ignore N-bases), at Amino-Acid level, the alphabet set includes 20 amino-acid types (Appendix .2, we ignore the newly discover Selenocysteine in this work).

Definition 3. A one-dimensional, finite automaton recognition process is depicted as a diagram (the word is '*tea*') $\boxed{\epsilon}$ -----> $\boxed{t}$ ⟶ $\boxed{te}$ ⟶ $\boxed{tea}$

A word of length k is a k-string of alphabets, presented by the union $\Lambda_\epsilon \cup \Lambda_1 \cup \Lambda_2 \cup \ldots \cup \Lambda_k$, for which $\Lambda_k \in \mathcal{S}(\Lambda)$, the symmetric group (power set) of Λ. Here, we adopt a stronger mathematical structure for all combinations of Λ as **group**. Specifically, the interactor $\circ$(operator) between two random elements a and b induces $a \circ b$ is also in Λ (closure). Besides, group structure allows associative $(a \circ b) \circ c = a \circ (b \circ c)$ and neutral existence (ϵ).

Definition 4. A language can be defined as a **set-former**, which is presented as

$$\{\boldsymbol{w} | \boldsymbol{w} \in \mathcal{C}\} \tag{5}$$

where $\boldsymbol{w}$ is the vector of word (ordered) and $\mathcal{C}$ is some **category** or **property**.

Example 1. The English alphabet consists of 26 distinct characters, which can be decomposed into two mutually exclusive sets: (1) vowels and (2) consonants. Let the set of property $\{\mathcal{C}_{\text{vowel}}, \mathcal{C}_{\text{consonant}}\}$ yields a set decomposition

$$\mathbb{B}_{\text{English}} = \mathbb{B}_{\text{vowel}} \times \mathbb{B}_{\text{consonant}}, \tag{6}$$

with $\mathbb{B}_{\text{vowel}} = \{\text{a, e, i, o, u}\}$ and $\mathbb{B}_{\text{consonant}} = \mathbb{B}_{\text{English}} \setminus \mathbb{B}_{\text{vowel}}$. Several source material states that two letters, Y and W, can function as consonants or vowels.

Example 2. AA-based genetic codes have a set of biological-induced properties $\{P_i\}$. Specifically, most of the amino acids have a chiral center with the α-carbon, except for *Glycine* (G) [23]. We can group *Glycine* with six other amino acids, including *Alanine*, *Valine*, *Leucine*, *Isoleucine*, *Methionine* and *Proline*; i.e.,
$\mathbb{B}_1 = \{G, A, V, L, I, M, P\}$ which are induced by property
$P_1 \coloneqq \{c_i | \text{Non}-\text{Polar, Non}-\text{aromatic Side Chain}\}$. The second and third sets of properties P_2 and P_3 are based on aromatic and polar side chains, respectively. Specifically, we have the corresponding subsets

- $P_1 = \{c_i | \text{Non}-\text{Polar, Non}-\text{aromatic Side Chains}\} \rightarrow B_1 = \{G, A, V, L, I, M, P\}$
- $P_2 = \{c_j | \text{Aromatic Side Chains}\} \rightarrow B_2 = \{W, F, Y\}$
- $P_3 = \{c_k | \text{Polar Side Chains}\} \rightarrow B_3 = \{S, T, N, Q, C\}$
- $P_4 = \{c_m | \mathbb{B}_{\text{AA}-\text{Genetic}} \setminus \{P_1 \cup P_2 \cup P_3\}\} \rightarrow B_4 = \{R, D, E, H, K\}$

2.2. Galois field

A special case of the ring $\mathbb{Z}/p\mathbb{Z}$ on integer modulo p is when p is a prime number, in which $\mathbb{Z}/p\mathbb{Z}$ is a field [24]. Another name for this mathematical structure is Galois prime field $\mathbb{Z}/p\mathbb{Z} \coloneqq \mathbb{F}_p$. This structure is used in cryptography or digital nets [25] but has not yet been adopted for word embedding of ML techniques. Of note, the field $\mathbb{F}_p$ has two properties: (1) the field does not contain proper subfields, and (2) This field structure is the proper subset of every finite field, i.e., $\mathbb{F}_p \subset \mathcal{F}$ for any $\mathcal{F}$ is finite fields [26]. The representation induced from $\mathbb{F}_p$ is

$$\begin{aligned} \psi\colon \mathbb{Z}/p\mathbb{Z} = \mathbb{F}_p &\rightarrow \mathbb{C}^* \\ m &\mapsto \psi[m] = \xi^m = e^{\frac{2\pi i m}{p}} \\ &= \cos\left(\frac{2\pi m}{p}\right) + i\sin\left(\frac{2\pi m}{p}\right) \\ m \in \mathbb{Z},\ m &\in \{0, 1, 2, \ldots, p-1\}, \end{aligned} \tag{7}$$

with $\mathbb{F}_p$ is cyclic, meaning the single generator can generate the entire group in $\{0, 1, \ldots, p-1\}$.

3. General method

3.1. Problem statement

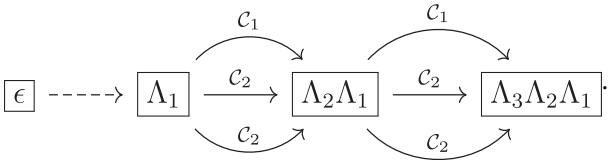
The primary aim of this research is to establish the mapping between the basis set of finite characters $\mathbb{B} = \{c_1, c_2, \ldots, c_n\}$ into quantum embedding space $\mathcal{V}_Q$ using group structure $\mathbb{Z}/n\mathbb{Z}$. Specifically, we aim to establish the maps

$$\begin{aligned} \text{Character Space} &\rightarrow \text{Quantum Feature Map} \rightarrow \text{Quantum Computer} \\ \mathbb{B} &\rightarrow \mathcal{V}_Q \rightarrow (\mathcal{X}, \mathcal{T}) \\ \boldsymbol{b} &\mapsto |\psi(\boldsymbol{b})\rangle \mapsto \boldsymbol{p}, \end{aligned} \tag{8}$$

where $(\mathcal{X}, \mathcal{T})$ are topological space of the quantum system.

3.2. Multi-dimensional finite automaton modeling

Definition 5. A multi-dimensional, finite automaton recognition process is depicted as a diagram.



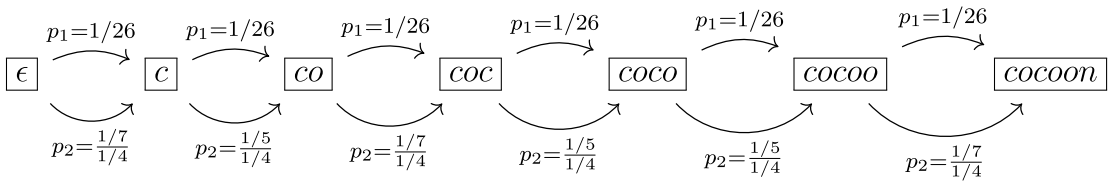
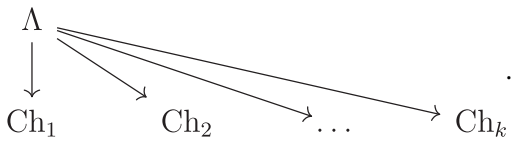
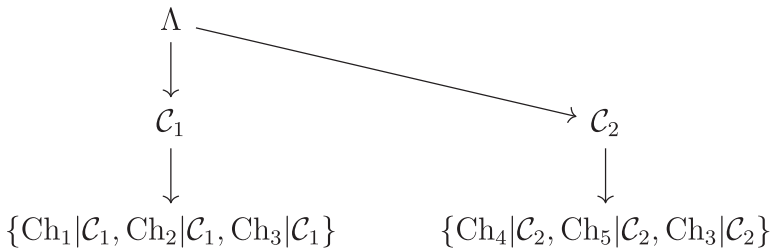


Figure 1. Finite automaton model with a probability of word transition for the word 'cocoon.'

To highlight the difference between our extension and the multi-dimensional automaton model, we will present a model in Def 3 and 5 as tree-based models.
Specifically, a one-dimensional automaton on k-alphabet set has the tree-based decomposition given as



In contrast, the same quantity of multi-dimensional, finite automaton modeling processes the hierarchical structure given as



We will give an example when the proposed multi-dimensional model shows advantages compared to uni-dimensional finite automaton

Example 3. We show the word 'cocoon' transition probability in the finite automaton model in figure 1. We denote the probability of sampling a character from Λ using 1D-model as p_1. Because we consider the word 'cocoon' in English vocabulary, which consists of 26 possibilities; thus, the given probability is $p_1 = 1/26$. From our multi-dimensional model, we partition the alphabet set into sub-structures by their categories, which yields conditional probability. For example, from the initial state of the model ϵ, we can have four categories of characters:

(i) $\mathcal{C}_{\text{vowel}} = \{\mathrm{a}, \mathrm{e}, \mathrm{i}, \mathrm{o}, \mathrm{u}\}$.

(ii) The set of consonant $\mathbb{B}_{\text{Consonant}}$ has non-prime order, which can be factored into the product of two prime 3×7. Thus, we can further perform set decomposition according to (case insensitive)

$$\begin{aligned}\mathcal{C}^1_{\text{consonant}} &:= \{c_i | c_i \in \{\mathrm{b}, \mathrm{c}, \mathrm{d}, \mathrm{f}, \mathrm{g}, \mathrm{h}, \mathrm{j}\}\}\\ \mathcal{C}^2_{\text{consonant}} &:= \{c_j | c_j \in \{\mathrm{k}, \mathrm{l}, \mathrm{m}, \mathrm{n}, \mathrm{p}, \mathrm{q}, \mathrm{r}\}\}\\ \mathcal{C}^3_{\text{consonant}} &:= \{c_k | c_k \in \{\mathrm{s}, \mathrm{t}, \mathrm{v}, \mathrm{w}, \mathrm{x}, \mathrm{y}, \mathrm{z}\}\},\end{aligned} \tag{9}$$

Hence, the conditional probability in the multi-dimensional automaton model is computed as

$$\mathbb{P}(\text{Character} = \text{“}c\text{”} | \mathcal{C}^1) = \frac{\mathbb{P}(\text{Ch} = c \cap \mathcal{C}^1)}{\mathbb{P}(\mathcal{C}^1)} = \frac{1/7}{1/4}$$

The advantage of the multi-dimensional model is to be equipped with the transition probability corresponding to the empirical distribution of vowels and consonants in the alphabet set ($5 \div 21$). In contrast, only naive transition probability is used in the one-dimensional model, which is equal for every character ($1/26$ in the English alphabet).

3.3. General protocol

We propose a new encoding method for Quantum Word Encoding (**QWE**) based on the mathematical structure of prime field $\mathbb{Z}/p\mathbb{Z}$. First, we transform the corpus set $\mathbb{B}$ into the category of sets (Def 4) (nested or hierarchical set) to reduce such space complexity, given by $\mathbb{B} = \{B_1, B_2, \ldots, B_k\}$ based on the properties $\{\mathcal{C}_1, \ldots, \mathcal{C}_k\}$ such that

C1. $B_k \coloneqq \{c_. | c_. \text{satisfy} P_k\}$

C2. $B_u \cap B_v = \varnothing, \forall\, u \neq v$

C3. $\bigcup_k B_k = \mathbb{B}$

As a result, the given problem in equation (8) becomes

$$\begin{aligned} &\text{Character Space} \rightarrow \text{Quantum Feature Map} \\ &B_1 \times B_2 \times \ldots \times B_k \rightarrow \mathcal{V}_Q^1 \times \mathcal{V}_Q^2 \times \ldots \mathcal{V}_Q^k \\ &(\boldsymbol{b}_1, \boldsymbol{b}_2, \ldots, \boldsymbol{b}_k) \mapsto (|\psi(\boldsymbol{b}_1)\rangle, |\psi(\boldsymbol{b}_2)\rangle, \ldots, |\psi(\boldsymbol{b}_k)\rangle) \end{aligned} \tag{10}$$

for each $b_i \in B_i$. By this decomposition, we can use the same ζ-basis for different subsets B_u and B_v. Specifically, if $|B_u| = |B_v| = p$, we can construct the complex-valued representations for B_u and B_v by the ring $\mathbb{Z}/p\mathbb{Z}$ on integer modulo p.

Protocol 1. Generalization: Given an n-character corpus $\mathbb{B} = \{c_1, c_2, \ldots, c_n\}$ with the set decomposition by property set $\{\mathcal{C}_1, \ldots, \mathcal{C}_k\}$ into $\mathbb{B} = \{B_1, \ldots, B_k\}$ that satisfies C1-C3, we compute the embedding for a word $\boldsymbol{w}$ of length m by Algorithm 1. The output of our algorithm includes two components, which are (1) positional encoding tensor Λ and (2) quantum embedding of word $\bar{\Xi}$. The latter quantity $\bar{\Xi}$ is computed based on the matrix multiplication of Ξ and its dual conjugate-transpose. The amplitude operator $|.|: z = a + bi \mapsto \sqrt{a^2 + b^2}$ transforms complex-valued functions into real values to accommodate classical computational models.

Algorithm 1. Algorithm for Quantum Word Embedding of Words from $\mathbb{Z}/n\mathbb{Z}$

Given: $\mathbb{B} = \{c_1, \ldots, c_n\}$, set decomposition $\mathbb{B} = \{B_1, \ldots, B_n\}$ correspondingto property set $\{\mathcal{C}_1, \ldots, \mathcal{C}_k\}$.
Ensure: $B_u \cap B_v = \varnothing$; $\bigcup_k B_k = \mathbb{B}$; $B_u \neq \varnothing\, \forall\, u \in \{1, \ldots, n\}$
Initialize: $\Xi_1 = \Xi_2 = \ldots = \Xi_k = \varnothing$. $\Lambda_i = \Lambda_j = \ldots = \Lambda_k = \varnothing$, where Λ is the positional encoding.
Input: a word $\boldsymbol{w}_m$ of length m, constructed by $\mathbb{B}$.
for character in $\boldsymbol{w}_m$:
  if character $\in B_i = \{c'_1, \ldots, c'_p\}$
    $\Lambda_i \leftarrow 1$
    $\Lambda_j \leftarrow 0,\ i \neq j$
    if character $= c'_u (u \in \{1, \ldots p\})$:
      $\Xi_i \leftarrow \xi^u = e^{\frac{2\pi i u}{p}}$
      $\Xi_j \leftarrow 0,\ i \neq j$
  $\bar{\Xi} = |\Xi^* \times \Xi|$
return Quantum Embedding Tensor: $[\Lambda, \bar{\Xi}]$.

4. Applications

4.1. Easy cases

4.1.1. QWE of english vocabular

In this use-case, we apply the proposed Quantum Word Embedding (QWE) into English vocabulary by Algorithm 1. The English alphabet consists of 26 distinct characters, which can be decomposed into two mutually exclusive sets: (1) vowels and (2) consonants. Let the set of property $\{P_{\text{vowel}}, P_{\text{consonant}}\}$ yields a set decomposition

$$\mathbb{B}_{\text{English}} = \mathbb{B}_{\text{vowel}} \times \mathbb{B}_{\text{consonant}}, \tag{11}$$

with $\mathbb{B}_{\text{vowel}} = \{\text{a, e, i, o, u}\}$ and $\mathbb{B}_{\text{consonant}} = \mathbb{B}_{\text{English}} \setminus \mathbb{B}_{\text{vowel}}$. We have $|\mathbb{B}_{\text{vowel}}| = 5$ and $|\mathbb{B}_{\text{consonant}}| = 21$, with the decomposition of the mapping in equation (10) into:

$$\begin{aligned}\mathbb{B}_{\text{vowel}} \times \mathbb{B}_{\text{consonant}} &\rightarrow \mathcal{V}_v \times \mathcal{V}_c \\ \boldsymbol{v} &\mapsto |\psi\rangle_v \\ \boldsymbol{c} &\mapsto |\psi\rangle_c \end{aligned} \tag{12}$$

Protocol 2. We map decomposed categories from the Character space into the corresponding prime field

$$\begin{aligned}\mathbb{B}_{\text{vowel}} \times \mathbb{B}^1_{\text{consonant}} \times \mathbb{B}^2_{\text{consonant}} \times \mathbb{B}^3_{\text{consonant}} &\rightarrow \mathbb{F}_5 \times \mathbb{F}_7 \times \mathbb{F}_7 \times \mathbb{F}_7 \\ (\boldsymbol{v}, \boldsymbol{c}_1, \boldsymbol{c}_2, \boldsymbol{c}_3) &\mapsto (\xi^i_{p=5}, \xi^j_{p=7}, \xi^k_{p=7}, \xi^l_{p=7}).\end{aligned} \tag{13}$$

$\xi_{p=5}$ is the basis of $\mathbb{F}_5$, given by

$$e^{\frac{2\pi i m}{5}} = \cos\left(\frac{2\pi m}{5}\right) + i \sin\left(\frac{2\pi m}{5}\right). \tag{14}$$

with character mappings of vowels given as

$$\{\{\text{a, e, i, u, o}\}\} \rightarrow k \in \{\{0, 1, 2, 3, 4\}\} \rightarrow \{\{e^{\frac{2\pi i k}{5}}\}\}^4_{k=0}, \tag{15}$$

where $\{\{a, b\}\}$ presents ordered set. of note, the transformation in equation (10) is the transformation of the tuple, while encoding in equation (13) concerns ordered set, or *dictionary*. Likewise, the consonant encoding is presented by mapping to the prime field of order 7, given by

$$\begin{aligned}\{\{\text{b, c, d, f, g, h, j}\}\} &\rightarrow \{\{0, 1, 2, 3, 4, 5, 6\}\} \rightarrow \{\{e^{\frac{2\pi i k}{7}}\}\}^6_{k=0} \\ \{\{\text{k, l, m, n, p, q, r}\}\} &\rightarrow \{\{0, 1, 2, 3, 4, 5, 6\}\} \rightarrow \{\{e^{\frac{2\pi i k}{7}}\}\}^6_{k=0} \\ \{\{\text{s, t, v, w, x, y, z}\}\} &\rightarrow \{\{0, 1, 2, 3, 4, 5, 6\}\} \rightarrow \{\{e^{\frac{2\pi i k}{7}}\}\}^6_{k=0}.\end{aligned} \tag{16}$$

4.1.2. QWE of Amino-acid-based genetic codes

We can trivially apply the QWE method of the English vocabulary to present Amino Acid-presented (AA-based) genetic codes. Specifically, the set of characters $\mathbb{B}_{\text{AA}-\text{Genetic}}$ is spanned by one-letter codes given in table A1. However, directly applying the proposed QWE for English vocab to AA-based genetic codes is not optimal due to redundancy. In particular, the mapping from $\mathbb{B}_{\text{English}} \rightarrow \mathbb{B}_{\text{AA}-\text{Genetic}}$ is not one-to-one mapping. This implementation causes computational redundancy since many characters such as (B, J, O, U, X, Z) are included in the loop of computation for Ξ-tensor in Algorithm 1 but not representing AA-based codes.

Protocol 3. We propose a non-trivial adaptation of Algorithm 1 for AA-based genetic codes by establishing a new set of properties $\{\mathcal{C}_i\}$, that will induce completely new set decomposition $\{B_i\}$. This proposed set decomposition auto-induces embedding of prime fields, given by

$$\begin{aligned}B_1 \times B_2 \times B_3 \times B_4 &\rightarrow \mathbb{F}_7 \times \mathbb{F}_3 \times \mathbb{F}_5 \times \mathbb{F}_5 \\ (\boldsymbol{AA}_1, \boldsymbol{AA}_2, \boldsymbol{AA}_3, \boldsymbol{AA}_4) &\mapsto (\xi^i_{p=7}, \xi^j_{p=3}, \xi^k_{p=5}, \xi^l_{p=5}).\end{aligned} \tag{17}$$

Thus, the embedding of AA-based genetic codes based on prime fields is given by

$$\begin{aligned}\{\{G, A, V, L, I, M, P\}\} &\rightarrow \{\{e^{\frac{2\pi i k}{7}}\}\}^6_{k=0} \\ \{\{W, F, Y\}\} &\rightarrow \{\{e^{\frac{2\pi i k}{3}}\}\}^2_{k=0} \\ \{\{S, T, N, Q, C\}\} &\rightarrow \{\{e^{\frac{2\pi i k}{5}}\}\}^4_{k=0} \\ \{\{R, D, E, H, K\}\} &\rightarrow \{\{e^{\frac{2\pi i k}{5}}\}\}^4_{k=0}\end{aligned} \tag{18}$$

4.2. Hard case: QWE of DNA-based genetic codes

We tackle a challenging problem involving DNA-based genetic codes, which is a special case for our algorithm as the number of DNA basis is four (4, non-prime), and unique prime decomposition of 4 is two even primes 2×2. Thus, this case study will give generalization for the proposed method for the alphabet with non-prime cardinality.

4.2.1. Two-state Lackadaisical encoding (TLE)

We observe that the base pair set that spans DNA-based genetic codes $\mathbb{B}_{\text{DNA}} = \{\boldsymbol{A}, \boldsymbol{T}, \boldsymbol{G}, \boldsymbol{C}\}$ can directly be mapped to quantum states over the Bloch sphere due to two factors: (1) small cardinality of four elements and (2) biological relations for set decomposition. Due to the first factor, we can encode the base pair set $\mathbb{B}$ into the 2-state quantum system, which has the basis set $\mathbb{E} = \{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ can be mapped as $\mathbb{B}_{\text{DNA}} \rightarrow \mathcal{V}_Q \coloneqq \mathcal{H}^2 \otimes \mathcal{H}^2, \{\boldsymbol{A}, \boldsymbol{T}, \boldsymbol{G}, \boldsymbol{C}\} \mapsto \{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$. The corresponding rotation of embedded quantum states is presented in table 1.

Table 1. A transformation to rotations TLE.

$\mathcal{R}_0$	Base pairs	Quantum states	θ_0	θ_1
	$\boldsymbol{A}$	$\|00\rangle$	$-\frac{\pi}{2}$	$-\frac{\pi}{2}$
	$\boldsymbol{T}(\boldsymbol{U})$	$\|01\rangle$	$-\frac{\pi}{2}$	$\frac{\pi}{2}$
	$\boldsymbol{G}$	$\|10\rangle$	$\frac{\pi}{2}$	$-\frac{\pi}{2}$
	$\boldsymbol{C}$	$\|11\rangle$	$\frac{\pi}{2}$	$\frac{\pi}{2}$

We name this encoding scheme as Two-state Lackadaisical Encoding (TLE) since the basis $\{|ij\rangle;\ i, j \in \{0,\ 1\}\}$ can be trivially mapped to that of OHE by the tensor product $\otimes$; i.e.:

$$|00\rangle = |0\rangle \otimes |0\rangle = [1,0,\ 0,0]^\top,\ |01\rangle = |0\rangle \otimes |1\rangle = [0,\ 1,\ 0,\ 0]^\top,\ |10\rangle = |1\rangle \otimes |0\rangle = [0,0,\ 1,0]^\top,$$

and

$$|11\rangle = |1\rangle \otimes |1\rangle = [0,\ 0,\ 0,\ 1]^\top.$$

The TLE mapping is mathematically presented by: $\mathbb{B}_{\text{DNA}} \to \mathcal{V}_{\text{Q}} \to (\mathcal{X},\ \mathcal{T})$, $\{\{\boldsymbol{A},\ \boldsymbol{T},\ \boldsymbol{G},\ \boldsymbol{C}\}\} \mapsto |\psi(\boldsymbol{b})\rangle \mapsto (\theta_0,\ \theta_1)$ with rotations (θ_0, θ_1) given by R_0 in table 1.

4.2.2. Topological-cyclic encoding (TCE)

In this encoding method, we will leverage the biological relations of base pairs for the set decomposition of the proposed prime field encoding (Algorithm 1). We can present the set $\mathbb{B}$ as $\mathbb{Z}/4\mathbb{Z}$, which have only two non-trivial set decomposition, including:

S1. $\mathbb{B}_{\text{DNA}} = \{b_1,\ b_2\}$, with $|b_1| = |b_2| = 2$, in which b_1, b_2 satisfying C1-C3.

S2. $\mathbb{B}_{\text{DNA}} = \{\{\boldsymbol{b}\},\ \mathbb{B}_{\text{DNA}} \setminus \{\boldsymbol{b}\}\}$, with $|\{b\}| = 1,\ |\mathbb{B}_{\text{DNA}} \setminus \{b\}| = 3$.

We find that the set structure **S1** can represent the binding mechanism of base pairs, while the set structure **S2** can present a mutation mechanism. Specifically, we have

S1. In genetic code, $\boldsymbol{A}$ binds $\boldsymbol{T}$ and $\boldsymbol{C}$ binds $\boldsymbol{G}$. Thus, the set decomposition is induced by the property

$$P_{\text{Bind}} = \{c_i|\text{Binding Partner}\}, \tag{19}$$

given by $\mathbb{B}_{\text{DNA}} = \{\{\boldsymbol{A},\ \boldsymbol{T}\},\ \{\boldsymbol{C},\ \boldsymbol{G}\}\}$. We present the graphical abstract of such set decomposition in figure 2(A). Specifically, we present the binding relation in a solid line, whereas a dashed line presents the non-binding relation. In the most simple case, we can represent the set structure as $\mathbb{Z}/4\mathbb{Z} \equiv P_4$, a square in $2D$ space ($\mathcal{G}^{(2)}$). This is because of that there is no relation between $b_1 = \{\boldsymbol{A}, \boldsymbol{T}\}$ and $b_2 = \{\boldsymbol{C}, \boldsymbol{G}\}$. Therefore, we can present the geometric relations of two segments $\boldsymbol{AT}$ and $\boldsymbol{CG}$ as any relation such as parallel in figure 2(A) or orthogonal, disjointed in figure 2(B).

S2. Mutations are alterations from a base pair into other bases except itself. Thus, we have the set decomposition presenting mutation mechanism given as $\mathbb{B}_{\text{DNA}} = \{\{\bar{\boldsymbol{b}}\},\ \mathbb{B}\setminus\{\bar{\boldsymbol{b}}\}\}$. We present the graphical abstract of the set decomposition by a mutation in figure 2(B). We denote the base pair set by the ordered color set $\{\boldsymbol{A},\ \boldsymbol{T},\ \boldsymbol{G},\ \boldsymbol{C}\} \mapsto \{\text{black,blue,green,red}\}$ and the mutation relations are presented in white line within the triangles.

Protocol 4. There are four variations for 2-dimensional geometry (Figure 2(B)) that represents P_{Mutation}, which are

$$\begin{aligned}
\mathcal{G}^{(2)}_{\boldsymbol{A}} &= \{\boldsymbol{A}\} \mapsto \{\boldsymbol{T},\ \boldsymbol{G},\ \boldsymbol{C}\}\\
\mathcal{G}^{(2)}_{\boldsymbol{T}} &= \{\boldsymbol{T}\} \mapsto \{\boldsymbol{A},\ \boldsymbol{G},\ \boldsymbol{C}\}\\
\mathcal{G}^{(2)}_{\boldsymbol{G}} &= \{\boldsymbol{G}\} \mapsto \{\boldsymbol{A},\ \boldsymbol{T},\ \boldsymbol{C}\}\\
\mathcal{G}^{(2)}_{\boldsymbol{C}} &= \{\boldsymbol{C}\} \mapsto \{\boldsymbol{A},\ \boldsymbol{T},\ \boldsymbol{G}\},
\end{aligned} \tag{20}$$

Using a three-dimensional, regular tetrahedron $\mathcal{G}^{(3)}$ spanned by the quantum state of a symmetric, informationally complete, positive operator-valued measure (SIC-POVM) with the projection set $\{P_{\boldsymbol{A}}(.),\ P_{\boldsymbol{T}}(.),\ P_{\boldsymbol{G}}(.),\ P_{\boldsymbol{C}}(.)\}$ (figure 2(C)), we can represent the lower dimensional geometries $\{\mathcal{G}^{(2)}_{\boldsymbol{A}},\ \mathcal{G}^{(2)}_{\boldsymbol{T}},\ \mathcal{G}^{(2)}_{\boldsymbol{G}},\ \mathcal{G}^{(2)}_{\boldsymbol{C}}\}$ for mutating mechanisms. For example, we have $P_{\boldsymbol{A}}(\mathcal{G}^{(3)}) \mapsto \mathcal{G}^{(2)}_{\boldsymbol{A}}$, projecting $\boldsymbol{A}$ onto the central point of regular triangle $\Delta\{\boldsymbol{C},\ \boldsymbol{T},\ \boldsymbol{G}\}$. Similarly, we have

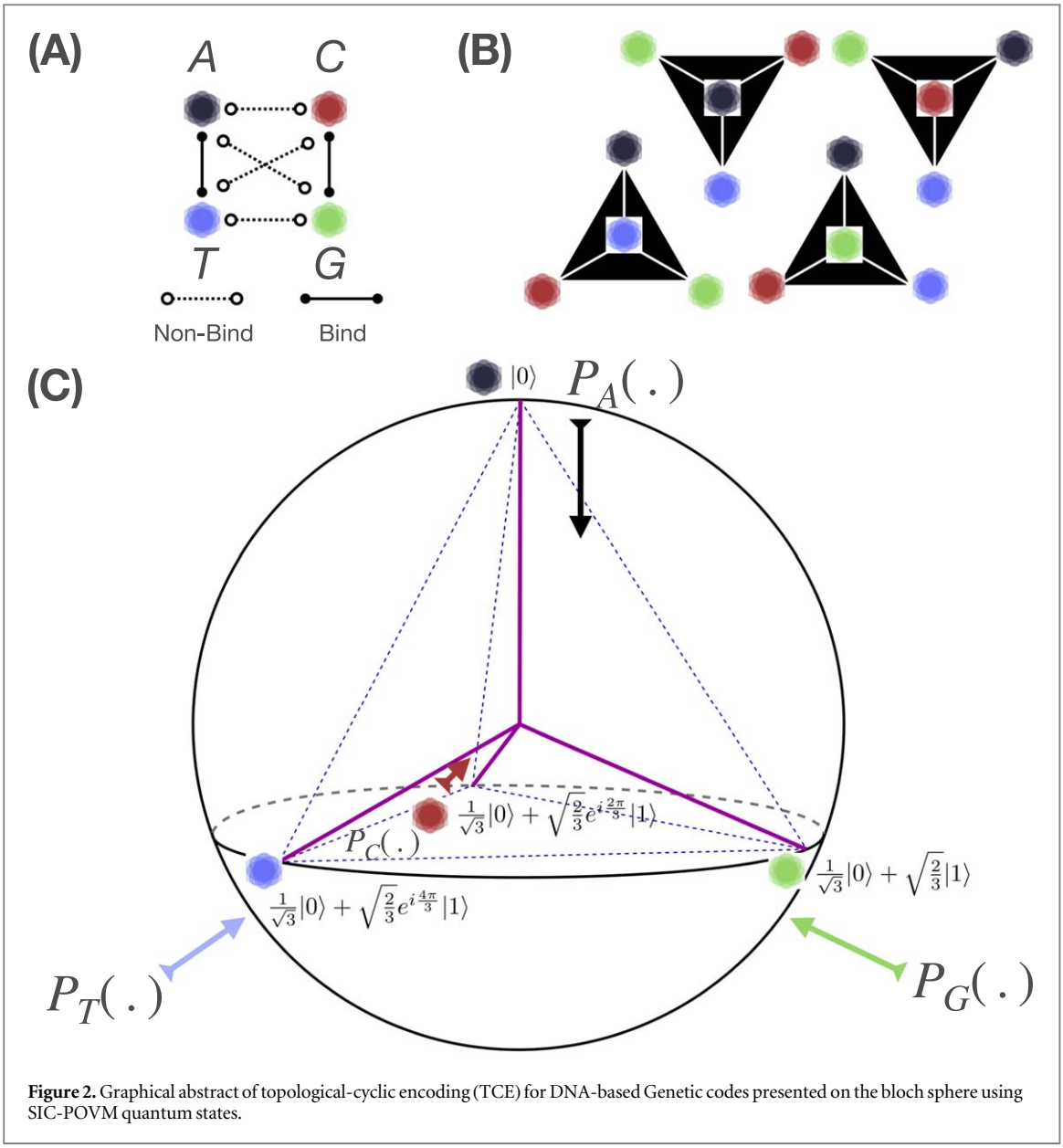


Figure 2. Graphical abstract of topological-cyclic encoding (TCE) for DNA-based Genetic codes presented on the bloch sphere using SIC-POVM quantum states.

$$\begin{aligned} P_A(\mathcal{G}^{(3)}) &\mapsto \mathcal{G}_A^{(2)} := \boldsymbol{A} \mapsto \Delta\{\boldsymbol{T},\ \boldsymbol{G},\ \boldsymbol{C}\} \\ P_T(\mathcal{G}^{(3)}) &\mapsto \mathcal{G}_T^{(2)} := \boldsymbol{T} \mapsto \Delta\{\boldsymbol{A},\ \boldsymbol{G},\ \boldsymbol{C}\} \\ P_G(\mathcal{G}^{(3)}) &\mapsto \mathcal{G}_G^{(2)} := \boldsymbol{G} \mapsto \Delta\{\boldsymbol{A},\ \boldsymbol{T},\ \boldsymbol{C}\} \\ P_G(\mathcal{G}^{(3)}) &\mapsto \mathcal{G}_C^{(2)} := \boldsymbol{C} \mapsto \Delta\{\boldsymbol{A},\ \boldsymbol{T},\ \boldsymbol{G}\}. \end{aligned} \tag{21}$$

We visualize the geometric projection of each base pair by a solid arrow with corresponding colors in **S2.** It is noted that the geometric abstract $\mathcal{G}^{(3)}$ also represents mutation mechanism $g(\bar{b}) \mapsto \mathbb{B}\backslash\{\bar{b}\}$, as

$$\begin{aligned} \boldsymbol{A} &\notin \Delta\{\boldsymbol{T},\ \boldsymbol{G},\ \boldsymbol{C}\} \\ \boldsymbol{T} &\notin \Delta\{\boldsymbol{A},\ \boldsymbol{G},\ \boldsymbol{C}\} \\ \boldsymbol{G} &\notin \Delta\{\boldsymbol{A},\ \boldsymbol{T},\ \boldsymbol{C}\} \\ \boldsymbol{C} &\notin \Delta\{\boldsymbol{A},\ \boldsymbol{T},\ \boldsymbol{G}\}. \end{aligned} \tag{22}$$

Thus, the SIC-POVM quantum states with the set of projections in figure 2(c) and equation (21) is adequate to present both bind and mutation mechanism of DNA codes. Notably, the encoding by projected geometry in equation (21) equivalent to finding the dual transformation

$$\begin{aligned} \mathbb{B} \to \mathbb{Z}/1\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \to \mathcal{V}_Q \\ \mathbb{Z}/1\mathbb{Z} \to \mathbb{Z}/3\mathbb{Z}, \end{aligned} \tag{23}$$

in which $\mathbb{Z}/1\mathbb{Z}$ is not a prime field since $p = 1$ is not a prime number.

Table 2. Context-dependent Permutation $\pi_{\boldsymbol{b}}$. If the encoded base pair is $\boldsymbol{b} = \zeta^{3/2}$, the its binding partner is ζ^0. $k(\boldsymbol{b})$ is parameter representing the size of biological molecular with mapping {{small,big }}:={{0, 1}}.

b	$k(b)$	π_b		
$\zeta^{3/2}$		ζ^{-1}	ζ^0	ζ^1
$\boldsymbol{A}$	1	$\boldsymbol{C}$	$\boldsymbol{T}$	$\boldsymbol{G}$
$\boldsymbol{T}$	0	$\boldsymbol{G}$	$\boldsymbol{A}$	$\boldsymbol{C}$
$\boldsymbol{G}$	1	$\boldsymbol{T}$	$\boldsymbol{C}$	$\boldsymbol{A}$
$\boldsymbol{C}$	0	$\boldsymbol{A}$	$\boldsymbol{G}$	$\boldsymbol{T}$

We propose a construction of the dual embeddings $(\bar{\phi}_{[m]},\ \phi_{[m]})$ by the extension of the prime field $\mathbb{F}_3 = \mathbb{Z}/3\mathbb{Z}$ with an additional rational element for $\mathbb{Z}/1\mathbb{Z}$. In particular, each vertex in a regular triangle Δ is isomorphic to the cyclic group $\mathbb{Z}/\mathbb{Z}_3$, with representations $\phi_{[m]}$ In particular, each vertex in a regular triangle Δ is isomorphic to the cyclic group $\mathbb{Z}/\mathbb{Z}_3$, with representations $\phi_{[m]}$

$$\begin{aligned}
&\mathbb{Z}/3\mathbb{Z} \to \mathbb{C}^* \\
&m \mapsto \phi_{[m]} = \exp\left(\frac{2\pi i m}{3}\right) \\
&m \in \{-1,\ 0,\ 1\} \in \mathbb{Z} \\
&\Xi = \{\zeta^{-1},\ \zeta^0,\ \zeta^1\}.
\end{aligned} \tag{24}$$

We define the duality function of $\phi_{[m]}$ as $\bar{\phi}_{[m]}$, given by

$$\begin{aligned}
&\mathbb{Z}/3\mathbb{Q} \to \mathbb{C}^* \\
&m \mapsto \phi_{[m]} = \exp\left(\frac{2\pi i m}{3}\right) \\
&m \in \left\{\frac{3}{2}\right\} \in \mathbb{Q}.
\end{aligned} \tag{25}$$

We have $\bar{\phi}_{[m]} = \exp\left(\frac{2\pi i}{3} \times \frac{3}{2}\right) = \exp(i\pi) = -1$. of note, the duality mapping $\bar{\phi}_{[m]}$ has only single value -1; i.e., the embedded vector space is collapsed from $\mathbb{C}^*$ into $\mathbb{R}$. To present embedding of geometry defined by the four base pairs in equation (21), we propose a context-dependent permutation of the set $\{\xi^{-1},\ \xi^0,\ \xi^1\}$

$$\eta(\boldsymbol{b}) = [\bar{\phi}_{[m]}^{k(b)}][\pi_{\boldsymbol{b}}[\zeta^{-1},\ \zeta^0,\ \zeta^1]] = [(-1)^{k(\boldsymbol{b})}][\pi_{\boldsymbol{b}}[\zeta^{-1},\ \zeta^0,\ \zeta^1]], \tag{26}$$

where $k(\boldsymbol{b})$ is defined as the characteristic function for each base pair. This function has only two values of -1 and 1. We propose a characteristic function based on the size of the molecular, with the transformation $\mathbb{B}_{\mathrm{DNA}} = \{\{\boldsymbol{A},\ \boldsymbol{T},\ \boldsymbol{G},\ \boldsymbol{C}\}\} = \{\{\text{big, small, big, small}\}\} = \{\{1,\ 0,\ 1,\ 0\}\}$. Besides, the permutation $\pi_{\boldsymbol{b}}$ allows the construction of distinctive ξ-basis set for each input base pair. Specifically, together with the proposed characteristic function $k(\boldsymbol{b})$, we construct the ξ-basis for DNA-based genetic codes in table 2.

The transformation in equation (20) is equivalent to

$$\begin{aligned}
&\{\boldsymbol{b}\} \mapsto \Delta\mathbb{B} \backslash \{\boldsymbol{b}\} \\
&\zeta^{3/2} \mapsto \Delta\Xi := \{\zeta^{-1},\ \zeta^0,\ \zeta^1\}
\end{aligned} \tag{27}$$

with $\zeta^{3/2} \notin \Xi$ since $\frac{3}{2} \notin \mathbb{Z}/3\mathbb{Z}$. We have the transformation $\boldsymbol{\eta}$ on the entire base pair set $\mathbb{B}$ is given by:

$$\begin{aligned}
\eta[\boldsymbol{A}] &= [(-1)^{k(\boldsymbol{A})}][\pi_A[\zeta^{-1},\ \zeta^0,\ \zeta^1]] \\
\eta[\boldsymbol{T}] &= [(-1)^{k(\boldsymbol{T})}][\pi_T[\zeta^{-1},\ \zeta^0,\ \zeta^1]] \\
\eta[\boldsymbol{G}] &= [(-1)^{k(\boldsymbol{G})}][\pi_G[\zeta^{-1},\ \zeta^0,\ \zeta^1]] \\
\eta[\boldsymbol{C}] &= [(-1)^{k(\boldsymbol{C})}][\pi_C[\zeta^{-1},\ \zeta^0,\ \zeta^1]]
\end{aligned} \tag{28}$$

The proposed η-function based on permutations of ξ-basis allows us to establish a distinctive set of bases to represent each base pair. Of note, these are only one of many ways to construct the mapping, as we have $3! = 6$ permutations of ξ-basis, >4 - the number of element in $\mathbb{B}_{\mathrm{DNA}}$.

To perform position encoding, we propose a convolutional operator applied to the given string code $\boldsymbol{x} = \{b_0,\ b_1,\ b_2,\ \ldots,\ b_m\}$, defined as

Table 3. Topological-Cyclic Encoding (TCE) based on quantum rotation gates.

Quantum Rotation ($\mathcal{R}_1$)	ζ^{-1}	ζ^{0}	ζ^{1}	Concatenate
σ_x	$R_{\sigma_x}(\zeta^{-1})_{2\times 2}$	$R_{\sigma_x}(\zeta^{0})_{2\times 2}$	$R_{\sigma_x}(\zeta^{1})_{2\times 2}$	$\Sigma_x[\zeta^{-1}, \zeta^{0}, \zeta^{1}]_{2\times 6}$
σ_y	$R_{\sigma_y}(\zeta^{-1})_{2\times 2}$	$R_{\sigma_y}(\zeta^{0})_{2\times 2}$	$R_{\sigma_y}(\zeta^{1})_{2\times 2}$	$\Sigma_y[\zeta^{-1}, \zeta^{0}, \zeta^{1}]_{2\times 6}$
σ_z	$R_{\sigma_z}(\zeta^{-1})_{2\times 2}$	$R_{\sigma_z}(\zeta^{0})_{2\times 2}$	$R_{\sigma_z}(\zeta^{1})_{2\times 2}$	$\Sigma_z[\zeta^{-1}, \zeta^{0}, \zeta^{1}]_{2\times 6}$
Concatenate	$R_{\sigma_{(.)}}(\zeta^{-1})_{6\times 2}$	$R_{\sigma_{(.)}}(\zeta^{0})_{6\times 2}$	$R_{\sigma_{(.)}}(\zeta^{1})_{6\times 2}$	$\Sigma_{[x,\,y,\,z]}[\zeta^{-1}, \zeta^{0}, \zeta^{1}]_{6\times 6}$

$$\boldsymbol{X} = \begin{bmatrix} [b_0, [b_1, b_2, b_3]] \\ [b_1, [b_2, b_3, b_4]] \\ \cdots \\ [b_{(m-3)}, [b_{(m-2)}, b_{(m-1)}, b_{(m)}] \end{bmatrix}_{(m-3)\times[1\times 3]}. \tag{29}$$

The encoding of $\boldsymbol{X}$, $\boldsymbol{\eta}[\boldsymbol{X}, \Lambda]$ is defined as

$$\boldsymbol{\eta}[\boldsymbol{X}, \Lambda] = \mathrm{diag}[(-1)^{k(b_i)}] \begin{bmatrix} \pi_{b_0}[\zeta^{-1}, \zeta^{0}, \zeta^{1}] \\ \pi_{b_1}[\zeta^{-1}, \zeta^{0}, \zeta^{1}] \\ \cdots \\ \pi_{b_{m-3}}[\zeta^{-1}, \zeta^{0}, \zeta^{1}] \end{bmatrix}$$
$$\boldsymbol{\eta}[\boldsymbol{X}, \Lambda]_{(m-3)\times 3} = \mathrm{diag}[\Lambda]_{(m-3)\times(m-3)}\, \Pi_{(m-3)\times 3}. \tag{30}$$

We construct the quantum feature maps by applying quantum rotation basis into the mapping $\boldsymbol{\eta}[\boldsymbol{X}, \Lambda]$. Of note, the representation $\boldsymbol{\eta}[\boldsymbol{X}, \Lambda]$ is a complex matrix, i.e., its entries is $\eta_{ij} \in \mathbb{C}$, which is associated with a rotation parameter

$$\theta_{ij} = \arg(\eta_{ij}). \tag{31}$$

As a result, we have the transformation to rotations

$$\begin{aligned} \Lambda \times \boldsymbol{\eta} &\to \mathcal{R}_1 = [-1, 1] \times M_{(n-3)\times 3}(\mathbb{R}) \\ (\Lambda, \boldsymbol{\eta}[\boldsymbol{X}, \Lambda]) &\mapsto (\Lambda, \Theta) \\ \eta_{ij} &\mapsto \theta = \arg(\eta_{ij}), \\ \Theta &= [\theta_{ij}](\text{real matrix}). \end{aligned} \tag{32}$$

Each rotation θ_{ij} can be encoded by parameterized quantum gates corresponding to Ox, Oy, Oz, given by

$$R_{\sigma_x}(\boldsymbol{\theta}) = \begin{bmatrix} \cos(\boldsymbol{\theta}/2) & -i\sin(\boldsymbol{\theta}/2) \\ -i\sin(\boldsymbol{\theta}/2) & \cos(\boldsymbol{\theta}/2) \end{bmatrix}, R_{\sigma_y}(\boldsymbol{\theta}) = \begin{bmatrix} \cos(\boldsymbol{\theta}/2) & -\sin(\boldsymbol{\theta}/2) \\ \sin(\boldsymbol{\theta}/2) & \cos(\boldsymbol{\theta}/2) \end{bmatrix}, R_{\sigma_z}(\boldsymbol{\theta}) = \begin{bmatrix} e^{-i\frac{\boldsymbol{\theta}}{2}} & 0 \\ 0 & e^{i\frac{\boldsymbol{\theta}}{2}} \end{bmatrix}.$$

Finally, we define the Topological-Cyclic Encoding (**TCE**) by quantum rotations in table 3, which can also be presented as

$$\begin{aligned} \mathcal{R}_1 &\to \Sigma_{[x,y,z]} \\ (\Lambda, \Theta_{(n-3)\times 3}) &\mapsto (\Lambda, \Sigma_{(n-3)\times[6\times 6]}) \end{aligned} \tag{33}$$

4.3. Dimensionality analysis

We have introduced three encoding methods for two sets of languages, including (1) English vocabulary and (2) genetic codes, applied to AA-based and DNA-based representations. We summarize these encoding schemes in table 4, with their corresponding embedded vector space.

5. Effectiveness analysis

5.1. Environmental setting

All experiments were carried out using Python 3.7.0, numpy 1.21.5, sci-kit-learn 1.0.2, and PyTorch 1.11 on an Intel i9 processor (2.3 GHz, eight cores), 16 GB DDR4 memory and GeForce GTX 1060 Mobile GPU with 6 GB memory.

5.2. Effectiveness analysis of QWE for english vocabulary

5.2.1. Comparison to baseline Word2Vec

We compare our proposed QWE with the Word2Vec [27] method with a comparable setting for word embeddings. Specifically, we use $m \times m$ embedding for a word of m characters using Word2Vec. Our proposed QWE is a 2D tensor; thus, the Word2Vec-induced embeddings are reshaped into the same dimension of $m \times m$

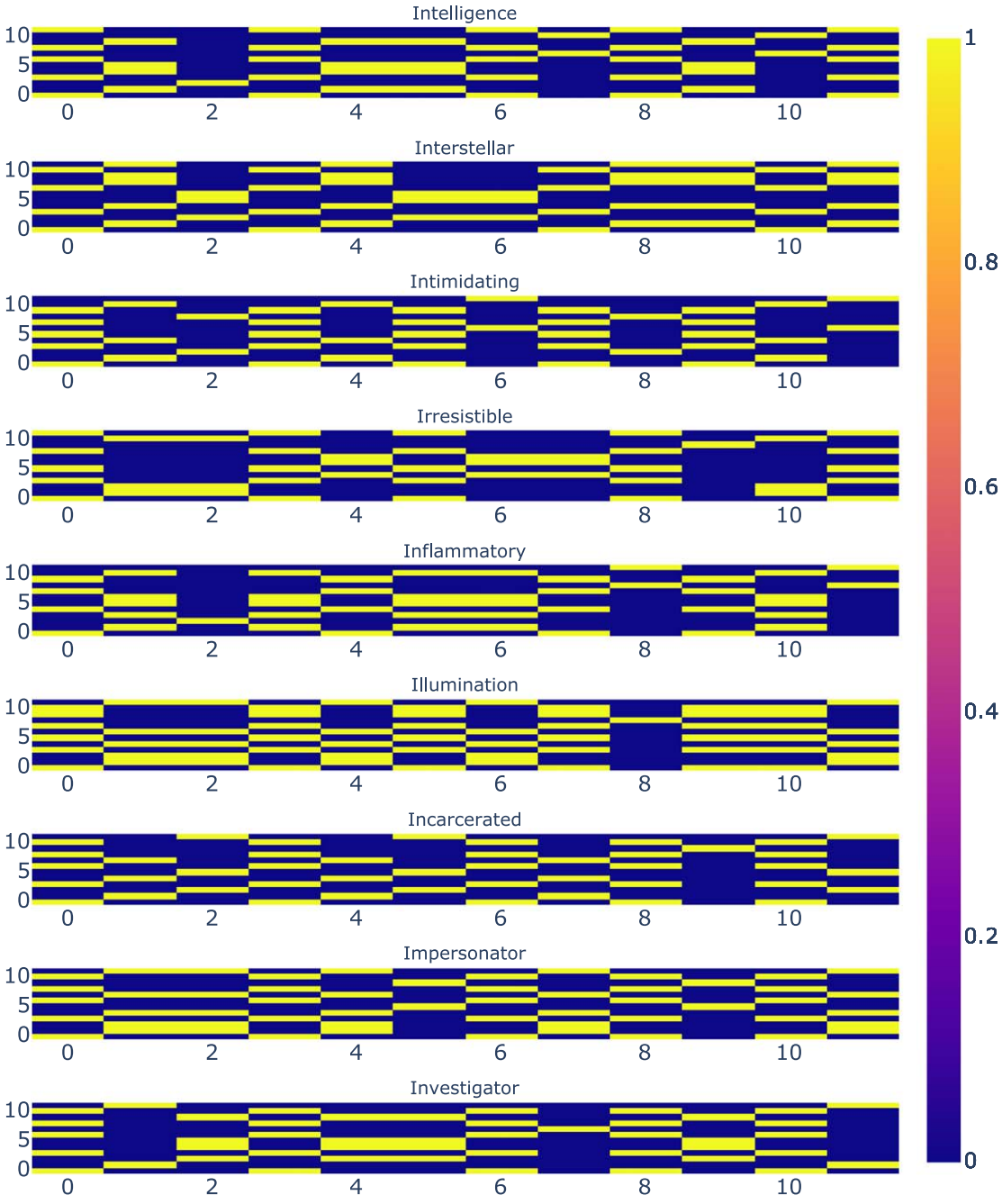


Figure 3. Feature Maps of 12-character English Vocabulary by Our Proposed QWE.

Table 4. Dimension comparison of different encoding scheme.

Encoding methods	Tensor dimension [batch size × sequence length × d_{model}]	Vector space
OHE (equation (1))	$[b \times m \times n_{\text{characters}}]$	$\mathbb{R}^4$
QWE (Algorithm 1)	$[b \times m \times m]$	$\mathbb{R}^m$
TLE (Table 1)	$[b \times m \times 2]$	$\mathcal{H}^2 \otimes \mathcal{H}^2$
$\mathcal{R}_0$	$[b \times m \times 2]$	$\mathbb{R}$
TCE (table 3)	$[b \times (m-3) \times [1 \times 3]]$	$\mathbb{C}$
$\mathcal{R}_1$	$[b \times (m-3) \times [1 \times 3]]$	$\mathbb{R}$
(Λ, Σ)	$[b \times (m-3) \times [1 \times [6 \times 6]]]$	$\mathbb{R}$

Table 5. Bench-marking corpora from http://vectors.nlpl.eu/repository/ used to benchmark the proposed QWE method. The word embedding vector is of dimension 300. - means not reported by the repository. **GSK**: Gensim Continuous Skip-gram, **GV**: Global Vectors—GloVe, **ELMo**: Embeddings from Language Models.

ID	Window	Corpus	Vocabulary size	algorithm	Lemmatization
1	10	British National Corpus	163, 473	GSK	✓
2	5	Norsk Aviskorpus/NoWaC	306, 943	GSK	✓
3	5	English Wikipedia Dump of February 2017	296, 630	GSK	✓
4	5	English Wikipedia Dump of February 2017	273, 992	GSK	✓
5	5	English Wikipedia Dump of February 2017	273, 930	GV	✓
6	5	English Wikipedia Dump of February 2017	273, 930	fastText	✓

(figure 3). We compare the two methods by measuring the mean of pair-wise Euclidean distance between word embeddings, given as

$$\text{Euclidean Distance} = \frac{1}{\text{number of words}} \sum (\boldsymbol{H}^{\boldsymbol{w}_1}_{m\times m} - \boldsymbol{H}^{\boldsymbol{w}_2}_{m\times m})^2 \tag{34}$$

Notably, good embeddings should be representative and well-stratified input sequences [28]; thus, the greater distance among embeddings can imply better encoding methods. For comparison, we use three corpora (batch size =300, 900, 1200) with vocabulary size ranging from $m = 6$ to $m = 13$, so 24 datasets are included in this analysis, showed in Figure 5.2 (**SuppMat**-S1). Based on the intrinsic evaluation (based only on extracted representations) by word distance, our proposed QWE outperforms the existing method Word2Vec since the evaluated score is above 0.2, with a mean distance of 0.3 recorded in all datasets. In contrast, such scores produced by Word2Vec are relatively small, close to 0 (**SuppMat**-S2). We find that the advantages of our methods are induced by using prime field encoding for word-specific encoding. Specifically, Word2Vec is corpus-wise embeddings, while our QWE is word-specific based on the character in the input vocabulary. Thus, the encoding vector produced by QWE could be more intrinsic and representative of input since each character within the word is encoded by different prime fields, which is intrinsic by the proposed set decomposition.

5.2.2. Comparison to bench-marking word embedding corpora

We compared the effectiveness of the proposed QWE to benchmarking word embedding corpora, summarized in table 5. We use six corpora from the Language Technology Group at the University of Oslo repository (http://vectors.nlpl.eu/repository/). The evaluation metric is the same as in equation (34). Since computing the pair-wise distance of word embeddings has $\mathcal{O}(n^2)$ complexity, we only randomly select 3, 000 words for each corpus and compute the embedding distance to the remaining vocabulary. The Gensim Continuous Skip-gram word embedding model trains word vectors by predicting the context words surrounding a given target word within a sliding window, optimizing for capturing syntactic and semantic relationships through efficient, scalable training on large corpora. Global Vectors for Word Representation (GloVe) is a word embedding technique that constructs word vectors by leveraging the statistical information of word co-occurrence in a corpus, capturing both global and local context to produce high-dimensional word representations. fastText word embedding is a technique that represents words as vectors by learning from character-level n-grams, capturing sub-word information, and allowing it to handle rare and out-of-vocabulary words effectively. Figure 5 (A)–(D) shows that the proposed QWE outperforms the continuous skipgram method, evaluated on four different corpora. In contrast, the proposed QWE is outperformed by global vectors and fastText method, depicted in figure 5 (E) and (F). QWE is not as good as GloVE, and fastText could be due to three main reasons:

(i) QWE is a non-training-based method (hard-coded), while GloVE and fastText are learning-based (variational coded). When comparing to more naive methods such as W2V or GSK, the proposed QWE is better with significant magnitude, as shown in figure 4 and 5(A)–(D).

(ii) GloVe and fastText embeds input sequence onto a fixed and high-dimensional vector space of 300 dimensions. In contrast, the QWE-induced vector depends on the number of characters in the input words, which has a much smaller dimensionality (see table 4).

(iii) GloVe and fastText are trained on extremely large corpora, which allows these methods to capture the global contextual dependencies between words. In contrast, QWE does not use such global dependency to encode words but uses only the component characters for encoding.

For further improvement of the proposed QWE, we can use GloVe and fastText methods on QWE-induced features.

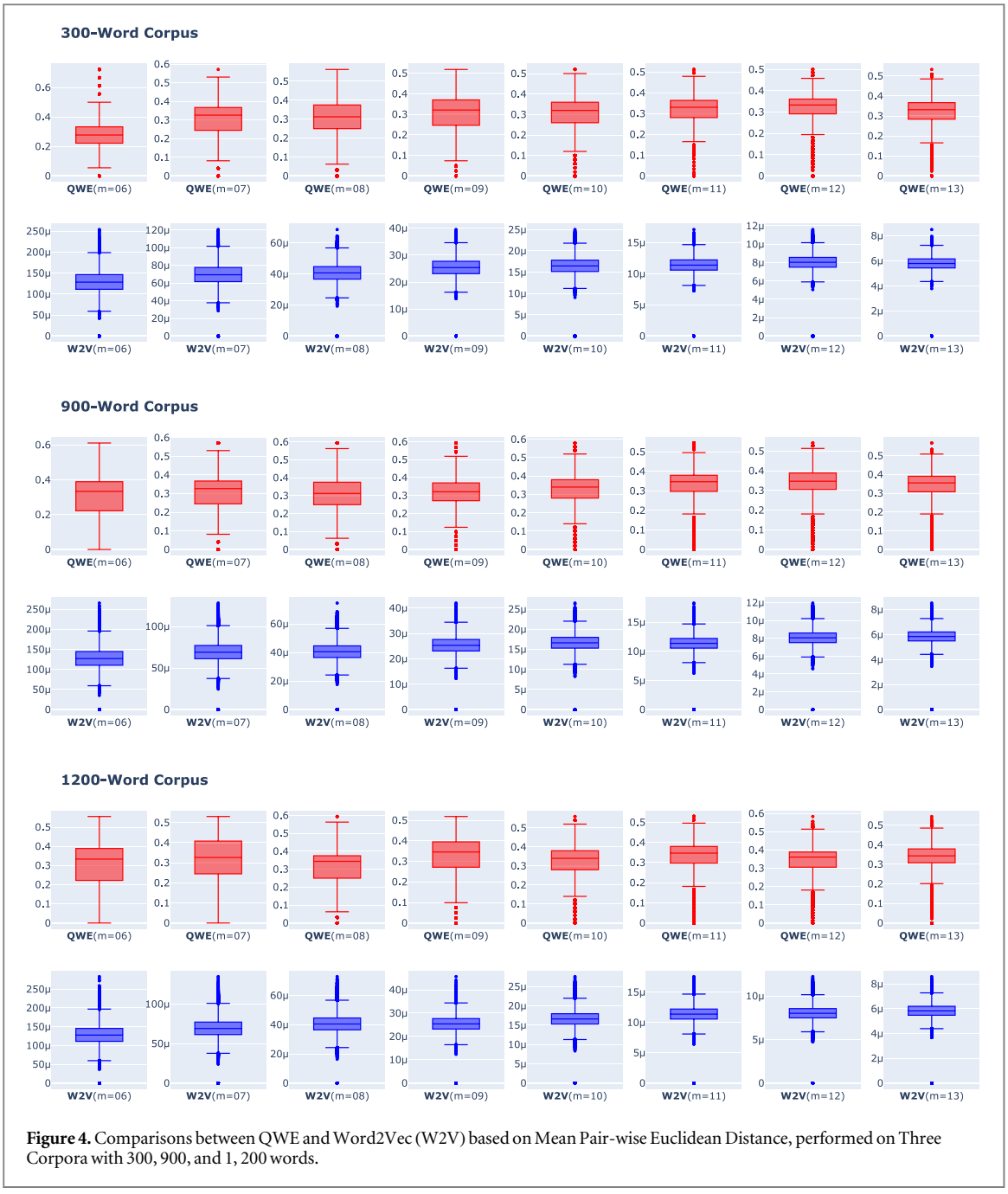


Figure 4. Comparisons between QWE and Word2Vec (W2V) based on Mean Pair-wise Euclidean Distance, performed on Three Corpora with 300, 900, and 1, 200 words.

5.3. Effectiveness analysis of QWE for AA-based genetic codes

In this use case, we demonstrate that the proposed QWE of AA-based genetic codes can predict RNA-protein interaction (RPI) based on the extracted features. Specifically, we use three large RPI databases, including RPI369 [29], RPI1807 [30], and RPI2241 [29], reported in figure 6. Of note, our proposed encoding technique has not addressed stop codon; thus, samples with such AA-based codes UAA (or TAA), UAG (or TAG), and UGA (or TGA) are excluded for further analysis. We establish the RPI score based on the distance between latent representations extracted by the QWE technique. This strategy is widely adopted in the related literature [31]. Here, we use two distance metrics based on (1) cosine similarity and (2) Mutual Information. Specifically, we compare the QWE of AA-based RNA of length m_1 with the QWE of AA-based protein sequence of length m_2, with $m_1 < m_2$ across all evaluated databases. We partition the targeted protein sequences into $(m_2 - m_1)$ sequences to compare QWE extracted by RNA-Protein pair with equal length. As a result, the final RPI score is computed by

$$\text{RPI score} = \text{argmin}\left\{\text{Latent Distance}(\text{QWE}_{\text{RNA}}, \text{QWE}_{\text{Protein}})\right\}_1^{m_2-m_1}, \tag{35}$$

To interpret this metric evaluation, the closer the latent QWE of AA-based genetic codes is, the higher the chance that RPI can be established.

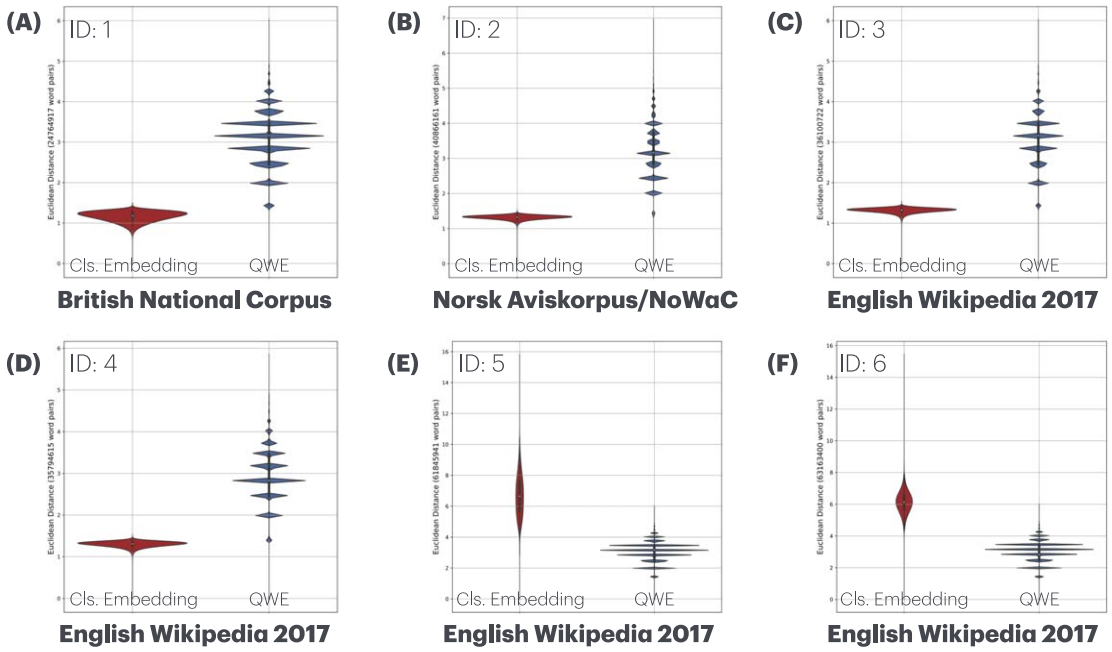


Figure 5. Comparison of QWE to bench-marking classical word embedding reported in table 5.

Based on the Mutual Information metric, our proposed QWE can efficiently and effectively predict RPI among two databases, RPI1807 and RPI2241, with a significantly small similarity between latent maps of binding partner and larger dissimilarity on a non-binding partner, with statistical significance level under $\alpha = 0.05$. In contrast, there is no statistical significance when comparing the same quantity in RPI369. Regarding the cosine similarity metric, our QWE only shows effectiveness in RPI1807 databases, while no statistical evidence is recorded from the remaining quantified datasets. Of note is that auto-encoders such as ANNs can be further applied to QWE to improve performance, which is worth investigating in future studies. To conclude, our QWE technique, a parameter-free encoding method, has substantial power in the prediction of RPI in two out of three databases (RPI1807 and RPI2241) based on the Mutual Information metric.

5.4. Effectiveness analysis of TCE for DNA-based genetic codes

We evaluate the relevancy [32] of two feature sets, including TLE and TCE, in the case study of gRNA analysis is the CRISPR-based gene editing technology (appendix .3). We show the result for TCE as TCE-(R_0), Lambda-(R_1) and Sigma-(R_1) based on the summary in table 4. To provide a comprehensive comparison, we perform evaluation using 15 databases reported in table A2, compared to 12 mathematical-driven features [33, 34] reported in table A3.

When considering the 23nt databases, we find that the proposed feature set has comparable Mutual Information with the target variable compared to OHE in all evaluated databases (SuppFig 1.1–1.13). In some data scenarios, such as datasets 1, 7, 12, 13, 14, and 16, our proposed feature set has slightly higher relevancy scores observed from feature group $570 - 720$. In some data cases, such as in the first database, nearly half of the features in the Σ-tensor are zero (SuppFig 1.1). Such sparse but representative features are a good property for embeddings since they could reduce computational resources for floating points, either for embeddings or the computational model applied to such embeddings. Non-zero features of TCE-Σ tensor have a significantly better relevancy score than conventional OHE. Besides, the tensor is computed by the product of a diagonal matrix (off-diagonal elements are zero) with a lower dimensional matrix based on permutation ($3(n-1)$ compared to $4n$ in OHE). Thus, the proposed Σ-tensor could have less floating point than an actual number of features ($720 = (23-3) \times 6 \times 6$ if further applying quantum rotation in table 3). Compared to MathFeature sets, our proposed encoding schemes perform similarly with these features, except in Dataset 1 (SuppFig 1.14). When comparing our proposed encoding to existing mathematical competitors, we observe the same results when considering longer input sequences (30nt databases). The proposed feature set is slightly better than OHE in databases 6, 7, 8, 9, and 16 (SuppFig 2.4–2.14). Besides, the proposed features are comparable with MathFeature sets, as shown in SuppFig 2.15–2.28.

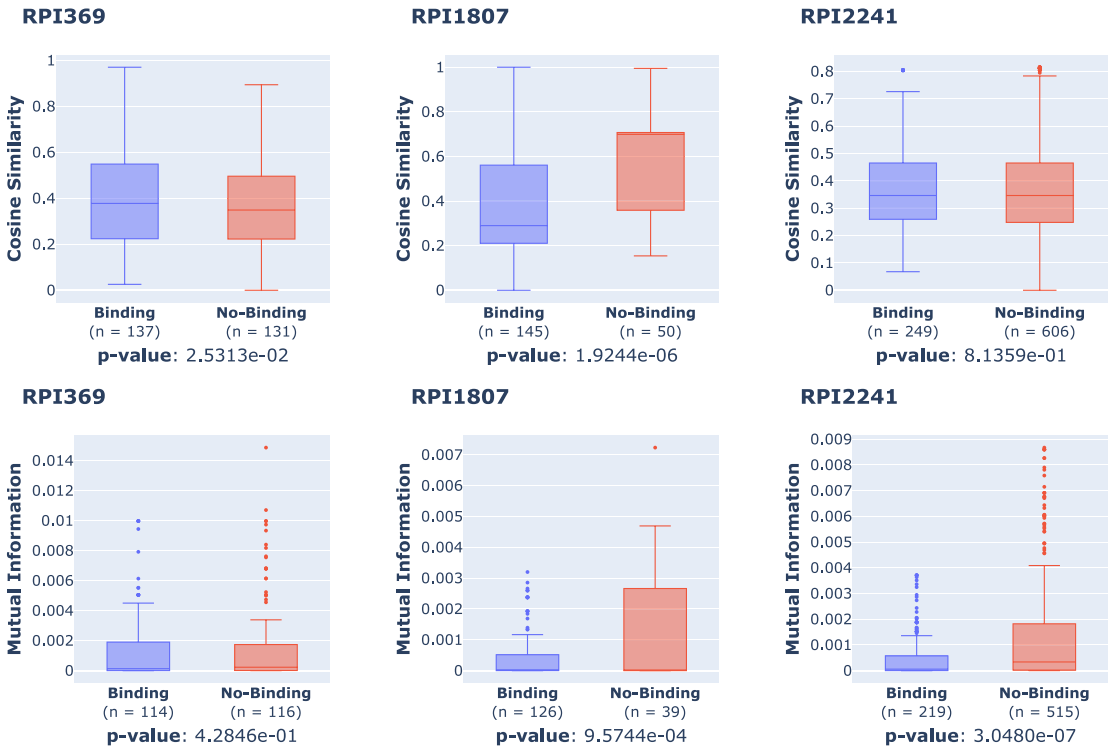


Figure 6. Effectiveness of Using QWE for Predicting RNA-Protein Interaction (RPI) based on the Distance of Latent Representations.

6. Conclusion

To this end, we have introduced the Quantum Word Embedding (QWE) technique for English vocabulary (Section 4.1.1) and AA-based genetic codes (section 4.1.2) with the generalization of the method discussed in section 3.3. Besides, we addressed DNA-based genetic codes by proposing TLE and TCE techniques, allowing information retrieval of the DNA codes over the topology of qubit Bloch sphere in section 4.2. We demonstrated the proof-of-concept for our proposed embedding techniques in three applications, including (1) information retrieval from the English vocabulary (Section 5.2), (2) using QWE of AA-based genetic code to predict RNA-Protein interaction (Section 5.3) and (3) gRNA analysis of the CRISPR-Cas 9 technology (section 5.4). In future work, we propose two research directions that can be extended from our studies, which are (1) to investigate the predictive power of the proposed feature set for ML/DL algorithm and (2) to quantify the effectiveness of such embeddings on actual, noisy quantum hardware.

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Data availability statement

No new data were created or analysed in this study.

Appendix

A1. Classical attention learning

The QSAN proposal addresses a significant limitation of self-attention-based models: their quadratic complexity. Specifically, the main goal of self-attention is to generate embeddings with global contextual information by learning the interaction between input tokens. This model construction is accomplished through the learning of three weight matrices. Queries $:=\boldsymbol{W}_Q \in \mathbb{R}^{n\times p}$, Keys $:=\boldsymbol{W}_K \in \mathbb{R}^{n\times p}$ and Values

Table A1. Conversion from 21 Amino Acids (AA) to One-character in English alphabet. Our analysis excludes the new 21st AA Selenocysteine(†).

Amino Acid	Letter	Amino Acid	Letter	Amino Acid	Letter	Amino Acid	Letter
Alanine	A	Glycine	G	Arginine	R	Histidine	H
Asparagine	N	Isoleucine	I	Aspartic acid	D	Leucine	L
Cysteine	C	Lysine	K	Glutamine	Q	Methionine	M
Glutamic acid	E	Phenylalanine	F	Proline	P	Serine	S
Threonine	T	Tryptophan	W	Tyrosine	Y	Valine	V

Table A2. Summary of databases used in our model evaluations and comparisons.

Index	Dataset	Number of Genetic codes	23-nt Seq.	30-nt Seq.
01	DeepCRISPR-CNN-CRNNCrispr [39]	16,749	✓	
02	DeepCas9 [40]	4,261		✓
03	DeepSpCas9 [41]	12,832	✓	✓
04[a]	CRISPRLearner (Chari-293t) [42]	1,234	✓	✓
05[a]	CRISPRLearner (Doench-hg) [42]	2,333	✓	✓
06[a]	CRISPRLearner (Doench-mel4) [42]	951	✓	✓
07[a]	CRISPRLearner (Moreno-Mateos) [42]	1,020	✓	✓
08[a]	CRISPRLearner (Wang Xu-hl60) [42]	2,076	✓	✓
09	OHE Average (all) [38]	23,665	✓	✓
10	A375 (Doench) [43]	2,333	✓	✓
11	HEK293T (Chari) [44]	1,234	✓	✓
12	HL60 (Wang-Xu) [45, 46]	2,076	✓	✓
13	HCT116 (Hart) [47, 48]	4,239	✓	✓
14	Azimuth [49]	5,310		✓
15	mEL4 (Doench) [43]	951	✓	✓

[a] Datasets with Pseudo-labels.

Table A3. Summary of feature set extracted by mathfeature [33, 34] Package.

Feature Name	Feature Dim. (23nt)	Feature Dim. (30nt)
Accumulated Nucleotide Frequency (ANF) + Fourier	[19 − 23]	[19 − 23]
Atomic Number	23	30
Binary	92	120
Classical Chaos Game Representation	46	60
Complex Number Encoding (C)	23	30
EIIP	23	30
Fourier Transformation	[19 − 23]	[19 − 23]
Integer Number Encoding (Z)	23	30
Entropy (Shannon-2mer)	2	2
Entropy (Shannon-3mer)	3	3
Real Number Encoding (R)	23	30
zCurve	69	90

$:= \boldsymbol{W}_V \in \mathbb{R}^{n \times p_v}$. The set of embedded representations is first projected onto each matrix to obtain

$$\begin{aligned} \boldsymbol{Q} &= \boldsymbol{X}\boldsymbol{W}_Q, \\ \boldsymbol{K} &= \boldsymbol{X}\boldsymbol{W}_K, \\ \boldsymbol{V} &= \boldsymbol{X}\boldsymbol{W}_V. \end{aligned} \tag{A1}$$

The output of self-attention layers, denoted as $\boldsymbol{Z} \in \mathbb{R}^{n \times p}$, is given as

$$\boldsymbol{Z} = \text{softmax}\left(\frac{\boldsymbol{QK}^T}{\sqrt{p}}\right)\boldsymbol{V}, \tag{A2}$$

equation (A1) states that the quadratic complexity arises because each token of the input sequence is attended to when computing the final embedding. Another way to interpret this limitation is via geometric deep learning (GLD [35]), which views the input sequence as a complete graph [36]. Thus, learning to embed the complete graph of n nodes also yields quadratic complexity.

The current approaches to attention learning face a challenge related to the need for position encoding. This means that the self-attention model does not have a sense of spatial information for each token within the input. Positional encoding (PE) is introduced through trigonometric functions [21] to address this issue. The purpose of this approach is to assist the model in learning the missing information related to spatial structure. However, using such numerical vectors can increase both the space complexity (in the case of non-parameter PE) and model complexity (in the case of variational PE), even though the embedding techniques can mitigate the limitations of self-attention models. It is worth noting that PE is an indispensable component for developing attention-based machine intelligence because models should capture sequential structures within input sequences.

A2. Character basis of amino acid-represented DNA codes

A3. CRISPR-Cas 9 and gRNA analysis

The use of single-guide RNA (gRNA) in genome engineering, specifically in the CRISPR/Cas9 system, has revolutionized the field of gene editing by allowing precise modifications to DNA sequences within cells [37]. The CRISPR/Cas9 system consists of two main components: the Cas9 enzyme, which acts as a molecular 'scissor' that cuts the DNA at a specific location, and the guide RNA (gRNA), which directs the Cas9 enzyme to the target site on the DNA. Computational approaches are typically used to design gRNAs to optimize the use of gRNA that achieve two objectives: (1) improving efficacy and (2) mitigating off-target cleavage. By evaluating the efficacy of gRNA design using these feature sets, we can gain insights into the utility of these features for improving the precision of gene editing techniques [38]. We summarize the evaluated databases in table A2 and compared information retrieval methods of DNA-based genetic codes in table A3.

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